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Time & Chronometry in the Works of Ibn al-Zarqālī of Toledo in al-Andalus during the 5th/11th Century

الوقت والتزمين في أعمال ابن الزرقالة الطليطلي

في الأندلس خلال القرن 5 هـ / 11 م

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Abstract:

Al-Andalus witnessed a remarkable scientific renaissance, particularly during the 5th century AH / 11th century CE, following the establishment of an astronomical school by Maslama al-Majrīṭī, who had revitalized the study of astronomy in al-Andalus a century earlier. Among the most prominent figures to emerge from this intellectual revival was Abū Ishāq Ibrāhīm ibn al-Zarqālī, also known as Arzachel in the West—a pioneer in applied astronomy and the construction of astronomical instruments, most notably the universal astrolabe (al-ṣafiha al-Zarqāliyya). His work also included the production of astronomical tables and calendars that became known in Europe as the Toledan Tables.

Ibn al-Zarqālī made groundbreaking contributions to the Andalusian astronomical renaissance. These included the measurement of the width of the Mediterranean Sea, precise calculation of the solar apogee, and laying the foundations of elliptical planetary orbits—concepts later developed by Copernicus and Kepler. His achievements earned him enduring recognition: Alfonso X of Castile ordered the translation of his works, the International Astronomical Union (IAU) named a lunar crater "Arzachel" (diameter: 96 km), and Spain issued a commemorative postage stamp bearing his image.

Keywords: Al-Andalus, Ibn al-Zarqālī, Astrolabe, Astronomical tables, Time & Chronometry.

الملخص:

عرفت الأندلس نهضة علمية كبيرة خاصة خلال القرن الخامس الهجري الحادي عشر ميلادي، بعدما أسس مسلمة المجريطي مدرسته الفلكية التي بعثت علم الفلك في الأندلس في القرن الرابع الهجري، كان من نتائجه ظهور "أبو إسحاق إبراهيم ابن الزرقالة" رائد علم الفلك التطبيقي وصناعة الآلات الفلكية (الإسطرلاب الكوني) أو ما يسمى بالصفحة الزرقالية، بالإضافة إلى صناعة الأزياج والتقاويم التي عرفت في أوروبا بالجدول الطليطية.

لقد ساهم ابن الزرقالة في النهضة الفلكية بالأندلس بأعمال رائدة مهدت لظهور نظريات حديثة في هذا المجال، كان من بينها قياس عرض البحر المتوسط، وقياس أوج الشمس بدقة، والتأسيس لنظرية إهليجية مدارات الكواكب التي اعتمدها كوبرنيكوس وكبلر فيما بعد.

إن الأعمال الرائدة التي قام بها ابن الزرقالة جعلت ألفونسو العاشر يأمر بترجمتها وأن يطلق الإتحاد الدولي لعلم الفلك (IAU) اسم فوهة قمرية على سطح القمر (Arzachel Crater) AL-Zarkali ذات القطر 96 كلم وكذلك إصدار طابع بريدي يحمل صورة ابن الزرقالة في إسبانيا.

الكلمات المفتاحية: الأندلس، ابن الزرقالة، الإسطرلاب، الأزياج، الوقت والتزمين.

1. INTRODUCTION

The Islamic-Arabic civilization witnessed significant advancements in rational sciences, especially after the stabilization of political and social conditions during the second century AH, under the Abbasid Caliphate in the East and the Umayyad rule in the Islamic West. This article highlights the achievements of astronomers in al-Andalus during the golden age of intellectual sciences in the 5th century AH (11th century CE), with a particular focus on the most distinguished figure of the era: Ibn al-Zarqālī. His pioneering works, still largely unexplored in Arabic sources, merit renewed scholarly attention.

Despite Ibn al-Zarqālī's recognized contributions to astronomy and instrument-making, his role in shaping the transition from Ptolemaic to early heliocentric models remains underexplored in modern scholarship, particularly regarding the transmission of his ideas to Europe and their influence on later astronomers such as Copernicus and Kepler.

While Ibn al-Zarqālī is widely recognized for his technical innovations in instrument-making and observational astronomy, his role in the theoretical transformation of medieval astronomy remains inadequately contextualized within the broader scientific milieu of al-Andalus. Specifically, the extent to which his work—particularly his proposals regarding elliptical planetary orbits, variable solar eccentricity, and universal instrumentation—challenged prevailing Ptolemaic models and facilitated the later transition to heliocentric cosmology in Europe has not been systematically examined through the dual lenses of scientific innovation and knowledge transmission.

This study seeks to investigate how Ibn al-Zarqālī's integrative approach to observation, instrument design, and theoretical critique contributed to an evolving astronomical paradigm in the 11th-century Islamic West, and to trace the pathways through which his ideas permeated later Eastern and Western astronomical traditions.

1. IBN AL-ZARQĀLĪ: LINEAGE, EARLY LIFE, AND ACHIEVEMENTS

1.1 Lineage and Early Life

Ibrāhīm ibn Yaḥyā al-Naqqāsh al-Tujībī al-Zarqālī, widely known as Ibn al-Zarqālī or simply al-Zarqālī, was also referred to by Western sources as "Arzachel." The geographer al-Gharnāṭī (d. 1137) called him "al-Zarqāl"¹ while others referred to him as "al-Zarqālī" or "Ibn al-Ṭulayṭulī"² The Andalusian judge Ṣā'id al-Andalusī called him "al-Zarqiyāl." His name is occasionally found written with a feminine ending (al-Zarqāla), as noted by Ibn al-Abbār³. Modern scholar Qadri Hafiz Tuḡan included him among the foremost astronomers of the 11th century CE, an era also known as the "age of al-Karakhī."⁴

Ibn al-Zarqālī was born in Toledo in 420 AH / 1029 CE—though some sources cite 419 AH.⁵ He was a prominent Andalusian astronomer and mathematician, considered one of the greatest in both theoretical and applied astronomy of his time. After the political upheaval in Toledo in 1085, during the reign of Alfonso VI of Castile, he relocated to Córdoba along with a number of scholars. His most important works will be explored in detail below.

1.2 Death

The exact date of Ibn al-Zarqālī's death remains a matter of scholarly debate. Some sources claim he died in 493 AH / 1100 CE⁶, while others specify Friday, the 8th of

Dhū al-Ḥijjah—the Day of Tarwiyah—in the same year (493 AH / 1099 CE)⁷. Yet another account places his death earlier, in 480 AH / 1087 CE, in Seville.

1.3 The Writings of Ibn al-Zarqālī

Before delving into his scientific accomplishments, it is useful to consider the definitions of astronomy (‘ilm al-hay’a) offered by some of the leading Muslim scholars:

Ibn Khaldūn defined astronomy as “the science that investigates the motions of the fixed and moving stars, and by analyzing the characteristics of these motions, deduces the shapes and configurations of the celestial spheres through geometrical methods.”⁸

Mu’ayyad al-Dīn al-‘Urḍī of Damascus (d. 1266 CE) stated: “The nobility of any science lies in either the nobility of its subject or the certainty of its proofs—or both. Astronomy excels in both: its subject is the most wondrous of God’s creations, and its proofs are based on definitive geometric and arithmetic reasoning.”⁹

Ibn Sīnā (Avicenna) (d. 428 AH / 1037 CE) wrote: “Astronomy is the science that examines the condition of the parts of the universe, their shapes, relative positions, measurements, and the distances between them, as well as the movements of the celestial spheres and the stars, using spheres, segments, and circles through which these motions are completed.”¹⁰

In the Field of Authorship:

Ibn al-Zarqālī authored numerous books and treatises. Among his most significant works are:

- Treatise on the Use of the Universal Astrolabe for All Latitudes
- The Book of Operations with the Zīj Plate (al-Ṣafīḥa al-Zījīyya)

- Introduction to the Science of Astrology (al-Madkhal fī ‘Ilm al-Tanjīm)
- The Book of Regulation (Kitāb al-Tadbīr)

The Most Important Works and Achievements of Ibn al-Zarqālī in Astronomy

It is nearly impossible to exhaustively enumerate the scientific contributions of Ibn al-Zarqālī in the field of astronomy—given its numerous subfields—within the limited scope of a short article. Nevertheless, his works can be systematically classified into six principal domains:

1. The construction of the astrolabe
2. Astronomical tables and calendars
3. Astronomical geography
4. Observational astronomy
5. Astronomical theory
6. Horology and the science of timekeeping

1. Astrolabe Construction

To begin, it is essential to understand the nature of the astrolabe as defined by some classical Islamic sources. Al-Bīrūnī, for example, described it as:

"An instrument whose name the Greeks called 'astrolabe,' meaning 'mirror of the stars'... Through this instrument, one can determine time and ascertain the passing of day and night with the least effort and greatest accuracy¹¹. It has both a visible and a hidden side, with various components unified around a central pivot, engraved with specific lines and names designed for measurement and identification." According to Ibn al-Nadīm, the first Muslim to construct an astrolabe was Ibrāhīm ibn Ḥabīb al-Fazārī (d. circa 180 AH / 796 CE), who built both planispheric and flat models.¹²

Origins and Influences of Andalusian Astronomy

Andalusian astronomy was profoundly shaped by Eastern (Mashriqi) traditions, particularly during the Umayyad period and the early Abbasid scientific renaissance. By the mid-9th century CE, however, al-Andalus had begun to emerge as a center of original astronomical production. Among the pioneers of this early phase was the polymath ‘Abbās ibn Firnās (d. 274 AH / 877 CE), known for his flight experiments in the palace of al-Ruṣāfa in Córdoba, the construction of a celestial dome in his home, and his invention of a mechanical celestial globe, which he gifted to ‘Abd al-Raḥmān II. He also designed a water clock capable of determining the Islamic prayer times even when celestial bodies were not visible—a remarkable feat for its time and gifted it to Prince Mohamed¹³.

The School of Maslama al-Majrīṭī (d. 1007 CE)

Maslama al-Majrīṭī is widely regarded as the true founder of Islamic astronomy in al-Andalus. His introduction of the Sindhind astronomical system—based on al-Khwārizmī’s version—during the reign of ‘Abd al-Raḥmān II proved to be pivotal. Together with his student Ibn al-Ṣaffār, al-Majrīṭī modified and adapted the Sindhind tables for Andalusian use.¹⁴

Ibn al-Zarqālī later affirmed al-Majrīṭī’s observational precision, stating that he had recorded the position of the star Regulus (Qalb al-Asad) in 368 AH / 979 CE with a longitudinal value of 135.40°. Furthermore, Ibn al-Zarqālī drew upon the works of earlier astronomers¹⁵, notably Ibn Mu‘ādh, who had relied on al-Khwārizmī to determine the solar apogee at 77.50° from the vernal equinox—a value that Ibn al-Zarqālī would later adopt in his treatise on the universal plate (al-Ṣafīḥa al-Jāmi‘a)¹⁶. Another precursor, Ibn al-Samḥ, had authored one of the earliest known works

on the construction of universal plates for planetary calibration, laying foundational groundwork for Ibn al-Zarqālī’s later innovations.

The real breakthrough in astrolabe design, however, occurred in the 11th century CE (5th century AH), primarily through the efforts of two astronomers: ‘Alī ibn Khalaf and Ibn al-Zarqālī.¹⁷

During the Taifa period, particularly in the 4th and 5th Islamic centuries, Andalusian astrolabes were typically equipped with multiple plates—each corresponding to the latitude of a specific city. It was Ibn al-Zarqālī who revolutionized this approach by developing the universal astrolabe, inaugurating a new era in astronomical instrumentation.¹⁸

In 440 AH / 1048–1049 CE, he authored a seminal treatise divided into one hundred chapters on an instrument he called the Ṣafīḥa ‘Abbādiyya (named after al-Mu‘tamid of Seville)¹⁹, also known in Latin sources as Spahener Oziacheris. This universal plate could be used across all latitudes and featured, on one side, a dual grid of equatorial and ecliptic coordinates along with a ruler for horizon calibration; the reverse side displayed a zodiacal scale, a stereographic projection of the celestial sphere, a sine quadrant, and a diagram of the moon’s orthogonal distance from Earth at any given time.²⁰

Ibn al-Zarqālī also gifted a second plate to al-Mu‘tamid, naming it the al-Shakkāziyya. In this sophisticated design—detailed in a sixty-chapter treatise—he replaced the equatorial projection with a stereographic projection along the solstitial colure (the celestial great circle passing through the solstices). This marked a genuine revolution in the design of astrolabes. Despite its mathematical elegance, the Shakkāziyya was somewhat

difficult to operate, particularly because it lacked a rotating rete (spider), which meant it did not emulate the apparent rotation of the celestial sphere.²¹

According to the Spanish historian of science Julio Samsó, the *Ṣafīḥa al-Zarqālīyya* was indeed an original invention, albeit initially complex and unintuitive. Ibn al-Zarqālī recognized this and revised it, simplifying its use and rebranding it as the *Shakkāziyya*.²²

Around the same time, the astronomer and pharmacist Abū al-Ḥasan ‘Alī Khalaf al-Shajjār (or al-Ṣaydalanī) constructed a cosmic astrolabe in 464 AH / 1071 CE. Starting from Ibn al-Zarqālī’s plate, he identified the absence of a rete and incorporated one into his instrument, making it significantly more user-friendly. He presented this refined device to al-Ma’mūn Dhī al-Nūn, ruler of Toledo (435–467 AH / 1043–1072 CE), and named it the *Ma’mūnī Astrolabe*. This tradition—of dedicating scientific instruments to rulers in pursuit of patronage—was common among medieval scholars.

Ibn al-Zarqālī and the Equatorium

One of Ibn al-Zarqālī’s most significant contributions to astronomical instrumentation was his invention of the Equatorium, or *Ṭabaq al-Manāṭiq*—an analog computing device designed to determine the positions of the sun, moon, and planets without resorting to manual calculation.

- This instrument was composed of two primary plates.
- The first plate depicted the celestial spheres of all the known planets.
- The second displayed the epicycles, enabling accurate representations of planetary orbits according to the Ptolemaic model.

Remarkably, Ibn al-Zarqālī portrayed the orbit of Mercury as elliptical, suggesting he may have understood the elliptical nature of planetary paths more than four centuries before Johannes Kepler (1571–1630 CE). He also added comparative rings to his Equatorium, differentiating it from traditional Ptolemaic designs.

The Content of the "Zarqālī Plate" Treatise

Ibrāhīm al-Zarqālī authored a comprehensive treatise on the workings of the universal astrolabe, known as the *Zarqālī Plate*. The treatise consists of one hundred chapters and functions as a practical guide outlining the instrument’s use, benefits, and diverse applications. He later produced an abridged version of this treatise, commonly referred to as the *Shakkāzī Treatise*, which retains all the essential technical details of the original comprehensive version.

In the preface, al-Zarqālī surveys various astronomical instruments, both shadow-based and optical, including flat sundials, quadrants, celestial spheres, astrolabes, armillary spheres, and dioptras. He also evaluates their observational capacities and highlights their varying degrees of precision and utility.

In a subsequent section, al-Zarqālī introduces his own invention: a single, universal plate in which all engraved markings are integrated into one surface, making it applicable for all latitudes and usable across all horizons. What sets his device apart is its ease of use and its ability to track both fast- and slow-moving celestial bodies. According to al-Zarqālī, his plate meets all the requirements for accurate astronomical observation, incorporating the necessary mathematical figures and diagrams. He described it as a dual-mode instrument: one version being a complete but simplified model with minimal markings,

and the other a more condensed variant for practical use.

At the beginning of the treatise, al-Zarqālī offers a meticulous description of the visible face of the plate. He writes:

“The first of its features is the circle inscribed with degree markings, which represents the meridian circle. The five-degree intervals are marked above, while the diameter extending from the suspension ring to the base of the plate signifies the celestial equator. The vertical diameter crossing it represents the observer’s horizon.”

On the reverse side, he notes:

“The first element here is the altitude and shadow circle. In its upper half are the altitude divisions, and in the lower half are the straight and inverted shadow scales.”²³

Influence of the Zarqālī Plate on Eastern and Western Scholars

1. Influence on Eastern (Islamic) Scholars:

Following its diffusion across the Islamic world, the Zarqālī Plate (also known as the Universal Astrolabe) gained recognition among astronomers such as the Andalusian polymath Abū al-Ṣalt Umayya ibn ‘Abd al-‘Azīz. However, due to its technical complexity, it was initially difficult to master. This led several scholars to compose explanatory treatises to elucidate its operation, including:

- Mūsā ibn Ibrāhīm al-Muṭṭabbib, in his work *Misbāḥ al-Ṭālib wa Munīr al-Muḥibb*.
- Aḥmad ibn ‘Umar al-Shādhilī, in his *Treatise on the Shakkāzī Zarqālī Plate*.
- Abū Muḥammad al-Khujandī (Hāmid ibn Khidr), inventor of the sextant, in his book *On the Use of the Zarqālī Plate*.
- Ibn al-Sarrāj al-Shāmī (Shihāb al-Dīn Aḥmad ibn Abī Bakr), who

produced an astrolabe modeled after the Andalusian Shakkāzī Plate, calling his version the Sarrajiyya.

2. Influence on Western Europe:

The renowned orientalist Carra de Vaux stated that the Alfonsine Tables—compiled under King Alfonso X in the 13th century—represented the pinnacle of Arab astronomical achievements.²⁴

Nicolaus Copernicus himself employed the Zarqālī Plate in his astronomical observations. A variant of the astrolabe, dubbed the Catholic Astrolabe, was modeled after al-Zarqālī’s original design.²⁵

Scholarly Opinions and Al-Zarqālī’s Legacy in Astronomy

1. Nicolaus Copernicus (1473–1543), the renowned Polish astronomer, frequently cited al-Zarqālī’s ideas, notably in his seminal work *On the Revolutions of the Celestial Spheres*.²⁶

2. Ibn al-Abbār praised him as "the unmatched scholar of his era in arithmetic, observation, and the construction of astronomical tables. No figure in al-Andalus since the Muslim conquest matched his intellectual acumen and mastery of astronomical instruments."²⁷

3. Maḥmūd ibn Muḥammad al-Mar‘ashī (known as Mirim Chalabī), in his Persian-language epistle dedicated to Sultan Bāyazīd Khān (d. 938 AH), described the Zarqālī Plate as “the most refined, complete, inclusive, lightweight, and accessible of instruments.”²⁸

4. The Zarqālī Plate remained in use among both Muslims and Europeans at the dawn of the European Renaissance. Copernicus notably incorporated it into his research, and its design directly inspired the Catholic Astrolabe.²⁹

5. German orientalist Sigrid Hunke emphasized that European astronomers

depended heavily on al-Zarqālī's Toledan Tables.

6. Ibn Bājja alluded to an indirect theoretical treatise critiquing Ptolemy's method for determining the apogee of Mercury.

7. Ibn al-Qayyim mentioned now-lost writings by al-Zarqālī in which he corrected Ptolemy's lunar model in his own handwriting.

8. Gerard of Cremona translated al-Zarqālī's works into Latin during the 12th century, facilitating their wide dissemination in Europe.

9. Regiomontanus (Johannes Müller) in the 15th century authored a treatise discussing the utility of the Zarqālī Plate.

10. Western astronomers frequently referred to al-Zarqālī as the "Father of Astronomical Science."

11. Spanish author Mariano Calvo stated that al-Zarqālī transformed Toledo into "the Greenwich of the Middle Ages."

12. Al-Zarqālī influenced numerous Islamic philosophers and astronomers, including Ibn Bājja, Ibn Ṭufayl, Ibn Rushd the Younger, Ibn al-Kammād, Ibn al-Hā'im al-Ishbīlī, and Nūr al-Dīn al-Bīṭrījī.

13. Ram Landau, in his book *Islam and the Arabs*, remarked:

"Astronomy was not limited to the Arabs of the East; it was also vibrant in the West. In al-Andalus, al-Zarqālī invented an astrolabe that gained fame and laid the foundation for an integrated astronomical legacy."³⁰

14. In his *Ta' rīkh al-Ḥukamā'* (History of the Wise), Jamal al-Dīn al-Qifṭī wrote:

"Ibrāhīm ibn Yaḥyā al-Naqqāsh, known as al-Zarqālī, was the most astute observer of planetary motion and celestial configurations of his time. His Zarqālī Plate—still widely used among astronomers—encapsulated the marvels of celestial motion in a remarkably concise design."³¹

15. According to Dr. 'Alī 'Abd Allāh al-Difā',

"Ibrāhīm al-Zarqālī reigned supreme among 11th-century astronomers. He harmonized theoretical knowledge with technical craftsmanship, earning the title 'al-Naqqāsh' for his early work as an engraver—a skill that deeply informed his passion for astronomical instrumentation and applied science."³²

II. IN THE FIELD OF ASTRONOMICAL TABLES AND ZĪJES

Ibn Khaldun defines the zīj as: "A mathematical science based on numerical laws concerning the motion of each planet, derived from the demonstrations of astronomy, accounting for their velocity, slowness, direct or retrograde motion, and other phenomena. These computations determine the positions of the planets in their orbits for any hypothesized moment in time, using the laws extracted from astronomical treatises."³³

Al-Khwārizmī similarly states: "A zīj is a book through which the movements of the planets are calculated and from which calendars are extracted—meaning the yearly ephemerides of the planets. In Persian, zīj (زيج) means 'string' or 'cord'³⁴, alluding to the mathematical precision involved."

Ibn al-Zarqālluh (known in Latin sources as Azarquiel) began his engagement with astronomy through the manufacture of astronomical instruments—a craft at which he excelled, earning him the title al-Naqqāsh (the engraver). In 1060 CE, he entered the service of Sa'īd al-Andalusī, the qadi (judge) of Toledo and a prominent mathematician. During this period, he not only constructed scientific instruments for fellow scholars but also underwent two years of rigorous study under Sa'īd's mentorship. Having mastered the foundational principles of hay'a

(astronomical science), Ibn al-Zarqālluh joined the scholarly team established by Sa'īd and eventually assumed its leadership after his teacher's death.

Ibn al-Zarqālluh became renowned for his Toledan Zīj, the product of a collaborative scholarly effort involving Muslim, Christian, and Jewish astronomers. This work drew extensively on earlier astronomical tables, such as those by Maslama al-Majrītī, Abū al-Qāsim ibn al-Ṣaffār, Ṣā'īd ibn Aḥmad al-Taghlibī (also known as Abū al-Qāsim al-Jiyānī), 'Alī ibn Khalaf, and others.³⁵

Around 462 AH / 1070 CE, Ibn al-Zarqālluh commenced his work on astronomical tables, which were subsequently translated into Latin between the late twelfth and early fourteenth centuries³⁶. These Toledan Tables exerted a profound influence on Western astronomical thought for centuries. In fact, the renowned Polish astronomer Nicolaus Copernicus utilized them in his *De Revolutionibus Orbium Coelestium*, explicitly citing both Azarquiel (Ibn al-Zarqālluh) and al-Battānī³⁷. The German Orientalist Sigrid Hunke has likewise affirmed the critical importance of Azarquiel's tables to generations of European astronomers.

These tables, known collectively as the Toledan Zīj, were based on empirical observations conducted in Toledo between 1061 and 1080 CE. They significantly influenced later Spanish astronomers, particularly in the compilation of the Alfonsine Tables, commissioned two centuries after Azarquiel's death by King Alfonso of Castile, who ordered the translation of all his works into Castilian.

Of particular note are the solar calendars Azarquiel produced, extant in both Arabic and Latin. These solar tables were the outcome of extensive observations in Toledo,

primarily aimed at simplifying the calculation of terrestrial longitudes using the Earth's orbit, a method inherited from Babylonian astronomy. More broadly, the function of zījēs (astronomical tables) was to facilitate three key aspects of practical astronomy:

- Determining the times of prayer and the qibla (direction of Mecca) from any location.
- Calculating the visibility of the lunar crescent (hilāl).
- Astrology and prognostication.

According to Julio Samsó, Azarquiel's influence extended to other zījēs authored by figures such as Ibn al-Kammād, Ibn al-Hā'im, and Ibn al-Raqqām. His ideas also left a lasting impact on Maghrebi astronomers, notably Ibn Ishāq al-Tūnisī and Ibn al-Bannā' al-Marrākushī.³⁸

III. IN THE FIELD OF ASTRONOMICAL GEOGRAPHY

Ibn al-Zarqālluh enjoyed considerable acclaim in the field of astronomical geography. He was the first to calculate the length of the Mediterranean Sea with remarkable precision, correcting both al-Khwārizmī's and Ptolemy's earlier estimates. Ptolemy had posited a value of 62 degrees, while Azarquiel's measurement—42 degrees—closely aligns with modern calculations.

IV. IN THE FIELD OF ASTRONOMICAL OBSERVATION

The theoretical foundations of Ibn al-Zarqālluh's observational work were based on the models of Ptolemy and al-Khwārizmī, both of which, in turn, relied on the Indian Sindhind tradition. However, he also relied heavily on his own observational data, gathered over more than thirty years.

Azarquiel devoted twenty-five years of his life to solar observation. Between 468 AH /

1075 CE and 473 AH / 1080 CE, he authored a treatise, now lost, known in Latin and Arabic sources as "On the Solar Year" or "The Comprehensive Treatise on the Sun." According to these secondary references, in the year 467 AH / 1075 CE, he recorded the solar apogee at a longitude of 85.49° . He was thus the first to propose that the solar apogee undergoes its own motion at a rate of one degree every 279 Julian years.³⁹

Azarquiel introduced a solar model with a variable eccentricity, a significant innovation in both the Islamic West and Latin Europe, persisting in influence until the time of Copernicus. He also calculated the inclination of the solar apogee relative to the fixed stars, determining the rate of change to be 4.12 arcminutes per year—remarkably close to the modern value of 8.11 arcminutes. Most of his observations were carried out in Toledo. In a series of over 402 experiments, he determined that the position of the solar apogee at sunrise equaled its position at sunset⁴⁰. According to al-Zīj al-Kāmil fī al-Ta'ālīm by Ibn al-Hā'im, Azarquiel observed the Moon for thirty-seven years—a claim confirmed by multiple sources.⁴¹

V. IN THE FIELD OF ASTRONOMICAL THEORY

Over the course of a career spanning more than four decades, Ibn al-Zarqālluh formulated a number of pioneering theories that stemmed directly from his continuous empirical work. Among his most significant contributions:

He was the first to posit that planetary orbits are elliptical. His insights would later influence Copernicus and Johannes Kepler, the latter of whom formalized the concept in his first law of planetary motion, which states that planetary orbits are elliptical, with the Sun at one of the two foci. Ibn al-Zarqālluh described the elliptical shape of Mercury's

orbit as "the form known as elliptical among those versed in the science of astronomy."⁴²

He was also among the earliest scholars to discuss the motion of the solar apogee relative to the fixed stars, estimating it at 12.04 arcseconds per year. The current value is 11.08 arcseconds, differing by only 0.96 arcseconds—a negligible variance in the context of the celestial sphere, equivalent to an error margin of 0.000266 degrees or 0.000000740 of a full circle (360°).

Azarquiel was the first to mention the precession of the equinoxes (*précision des équinoxes*), an insight of great importance to later astronomical models.⁴³

Ibn Bājja (Avempace) references a now-lost theoretical work by Azarquiel titled *A Treatise Refuting Ptolemy's Method for Determining the Maximum Distance of Mercury*.

Azarquiel's decades-long lunar observations culminated in several crucial discoveries, including his formulation of the eclipse deviation theory (*inkhirāf al-kusūf*), pertaining to the solar eclipse.

VI. TIMEKEEPING AND HOROLOGY IN AL-ANDALUS: THE ANDALUSIAN WATER CLOCK

The Legacy of Ibn al-Zarqālī in Horology and Astronomy

Ibn al-Zarqālī, widely recognized for his expertise in theoretical and applied astronomy, also demonstrated remarkable skill in the field of horology—specifically, in the construction of mechanical timekeeping devices. Among his most celebrated achievements were two large-scale water clocks installed in the city of Toledo (*Ṭulayṭila*), known locally as *al-Baylatayn*. These were situated on the banks of the Tagus River near *Bāb al-Dabbāghīn* (The Tanners' Gate).⁴⁴

The functioning of these clocks was based on the lunar calendar, operating on a 29-day cycle in accordance with the phases of the moon. Their internal mechanisms were hydraulic, with each clock emptying completely at the end of every lunar month, only to be refilled gradually as the moon passed through its various phases—crescent, first quarter, gibbous, full, waning gibbous, last quarter, and back to crescent—effectively mirroring the 28 lunar mansions (*manāzil al-qamar*). These sophisticated devices remained in operation until the fall of Toledo in 478 AH / 1085 CE. One of the clocks continued functioning until 528 AH. King Alfonso VI of León and Castile, intrigued by the mysterious technology, commissioned a Jewish scholar named Hāmīs ibn Zubārah to decipher the operating principle of the first clock—an artifact that had astonished both the king and the intellectuals of the period. Despite his efforts, Hāmīs failed to unlock its secret. Nonetheless, the second clock continued to function for another fifty years. Both clocks were renowned for their astonishing precision—so much so that later generations compared them to modern timepieces.

CONCLUSION

The achievements of Muslim scholars such as Ibn al-Zarqālī stand as irrefutable evidence that the scientific awakening in al-Andalus was not merely a continuation of Greek heritage, as some Eurocentric narratives suggest, but rather an original and innovative movement. The astronomical renaissance of the 5th Islamic century (11th century CE) found in Ibn al-Zarqālī one of its foremost pioneers, especially in practical applications and the manufacture of astronomical instruments. Among his most groundbreaking contributions was his transformation of the

astrolabe from a regional instrument into a universal one—an invention that became known as the universal astrolabe (*al-aṣṭurlāb al-kawnī*). His creation of the al-Zarqālī plate (*al-ṣafīḥa al-zarqāliyya*) represented a paradigm shift in astronomical instrumentation, remaining in use well into the 16th century, particularly in maritime navigation and geographical exploration.

In the realm of astronomical theory, Ibn al-Zarqālī investigated the motion of fixed stars, the oscillation of the ecliptic's inclination, and the displacement of the centers of lunar and solar orbits. His extensive research into planetary motion notably anticipated the concept of elliptical orbits, thus laying the groundwork for Copernican heliocentrism and Kepler's laws of planetary motion.

He compiled the results of over twenty-five years of solar observations in a comprehensive treatise titled *al-Risāla al-Jāmi'a fī al-Shams* ("The Comprehensive Epistle on the Sun"). Moreover, he led a research team that produced highly accurate astronomical tables (*al-zīj*), which had a significant influence on European astronomical research. These tables were later translated into Castilian and Catalan under the directive of Alfonso X, and also into Hebrew.

Ibn al-Zarqālī's contributions to horology in Toledo culminated in the invention of a remarkable lunar water clock—an engineering mystery that remains unsolved to this day. He authored several works, of which four survive: *Kitāb al-Tadbīr* (The Book of Administration), *Kitāb al-'Amal bi-l-Ṣafīḥa al-Zījīyya* (On the Use of the Zīj Plate), *al-Madkhal fī 'Ilm al-Tanjīm* (Introduction to the Science of Astrology), and a treatise on the usage of a multipurpose plate adaptable to various latitudes.

Ibn al-Zarqālī's name has continued to be honored by the scientific community. In 1935, the International Astronomical Union (IAU) named a lunar crater on the moon's far side Arzachel Crater in his memory.

May God have mercy on Ibn al-Zarqālī for the knowledge he bequeathed to humanity.

This study has demonstrated that Ibn al-Zarqālī's work was not merely technical but theoretically innovative, particularly in his early proposal of elliptical planetary orbits and his precise solar observations. His influence extended beyond al-Andalus through Latin and Hebrew translations, directly informing Copernican and Keplerian astronomy. The research underscores the need to revisit Arabic astronomical manuscripts for earlier traces of heliocentric reasoning. Future studies could compare his solar model with contemporaneous Eastern Islamic astronomers to better map the circulation of ideas in medieval Eurasia.

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